

CHECK WHAT YOU LEARNT

Check Yourself - Unit 3**A. Choose the correct answer**

1. An ultrasonic sensor measures distance by
(a) seeing light (b) timing an echo (c) touching the object (d) weighing it
2. The distance formula divides by 2 because
(a) it is easier (b) sound travels there and back (c) the sensor is slow (d) of the resistor
3. The TRIG pulse must be held HIGH for about
(a) 2 microseconds (b) 10 microseconds (c) 10 milliseconds (d) 1 second
4. Every reading comes back as 0. The likely fault is
(a) a dead battery (b) TRIG and ECHO swapped (c) a missing LED (d) wrong baud rate
5. If the stand is 180 cm and the sensor reads 42 cm, the person is
(a) 222 cm (b) 138 cm (c) 42 cm (d) 90 cm
6. Averaging five readings is done to reduce
(a) voltage (b) noise (c) current (d) the delay

B. Fill in the blanks

- 1 Sound travels about _____ cm in one microsecond.
- 2 Height equals standHeight _____ the measured gap.
- 3 A number worked out from another measurement is called a _____ quantity.
- 4 Hair scatters the echo, which makes a person read too _____.

C. Answer in one or two lines

- 1 An echo returns in 900 microseconds. Work out the distance, showing your steps.
- 2 Write down your machine's largest error and one honest reason for it.
- 3 Explain why averaging makes a reading steadier.
- 4 Why is the sensor called a digital sensor when distance is an analog quantity?

UNIT 4

Intermediate Project: Smart Contactless Dustbin

Sessions 9, 10 and 11

Until now your machines have only reported. This one *moves*. A servo lifts a real lid when a hand comes near, holds it, and closes it again - and getting that to feel natural is harder than the code suggests.

By the end of this unit you will be able to:

- Explain how a servo differs from an ordinary motor.
- Design a mechanical linkage that matches the servo's travel.
- Combine a sensor and an actuator into one automatic machine.
- Identify false triggers and remove them.

SESSION 9

one class

Servo Motors and the Lid Mechanism

Today you will

- Learn how a servo knows where it is.
- Test the servo's travel before building anything.
- Design the bin and lid on paper.

An ordinary motor spins as long as you give it power, and it has no idea where it is. A **servo** is different. Inside its plastic case there is a small motor, a gearbox, and - most importantly - a sensor watching the output shaft. You tell the servo an angle, and its own internal circuit keeps adjusting until the shaft is at that angle, then holds it there against the weight of whatever is attached.

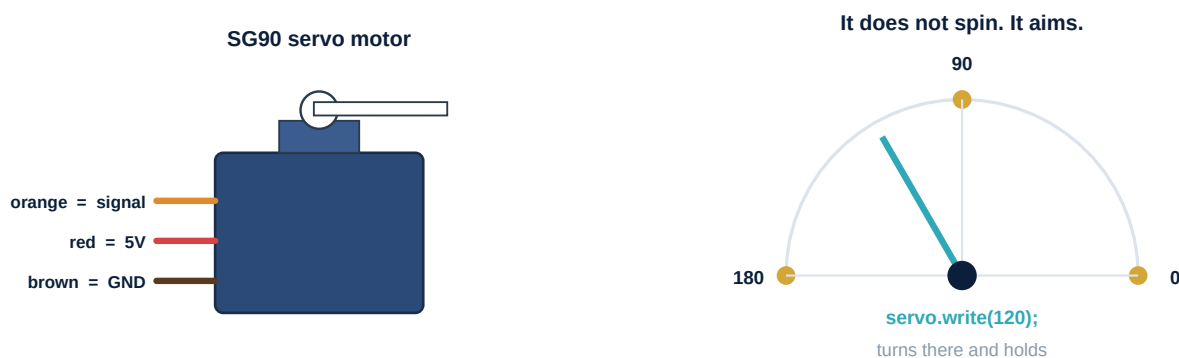


Figure 5.3 - A servo turns to an exact angle between 0 and 180 degrees

Servo wire	Colour on an SG90	Connect to
Signal	Orange or yellow	any digital pin, D6 in this project
Power	Red	5V
Ground	Brown or black	GND, shared with the Arduino

ServoTest.ino - run this before you build anything

```
#include <Servo.h>           // borrow the ready-made servo instructions

Servo lid;

void setup() {
  lid.attach(6);             // the servo's signal wire is on D6
  lid.write(0);              // start closed
}

void loop() {
  lid.write(90);             // open
  delay(2000);
  lid.write(0);             // close
  delay(2000);
}
```

Watch Out

Never force a servo horn round with your fingers. The gears inside are small plastic teeth and they strip easily - and a stripped servo cannot be repaired. Always move it by sending it an angle.

Try This

Before building the bin, run ServoTest and watch the horn carefully. Mark on paper exactly where the arm sits at 0 degrees and at 90 degrees. Your whole lid mechanism has to fit inside that sweep - design around the servo, not the other way round.

SESSION 10

one class

Building and Coding the Dustbin**Today you will**

- Build the bin, hinge the lid and mount the servo.
- Wire the sensor, servo, buzzer and lamp.
- Get the full open-hold-close cycle running.

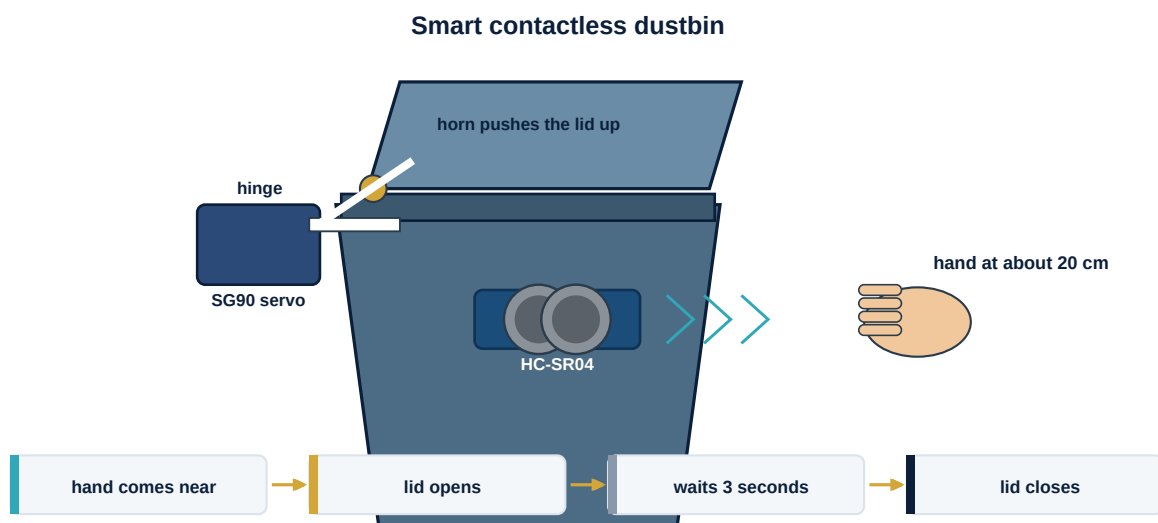


Figure 4.1 - A sensor, a motor, and a lid nobody has to touch

- Keep the lid **light**. A heavy cardboard lid makes the servo strain, buzz and eventually give up.
- Fix the servo body firmly to the bin. If the body moves instead of the horn, nothing works.
- Mount the sensor on the **front face**, pointing outwards and slightly upwards, at about the height a hand approaches.
- Make sure the open lid never blocks the sensor's view - a classic and very confusing fault.

Smart dustbin - connections

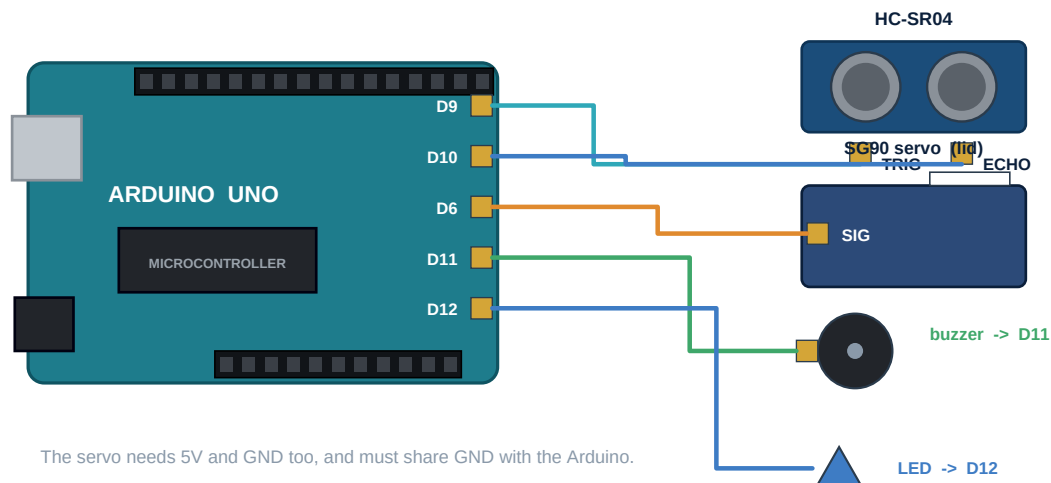


Figure 4.2 - One sensor in, one motor and two alerts out

Component pin	Arduino pin	Why
HC-SR04 TRIG	D9	The command to fire
HC-SR04 ECHO	D10	The pulse you time
Servo signal (orange)	D6	Lid angle
Servo red / brown	5V / GND	Power. GND must be shared with the Arduino
Buzzer +	D11	A short chirp as the lid opens
Blue LED long leg	D12 through 220 ohm	Shows the lid is open

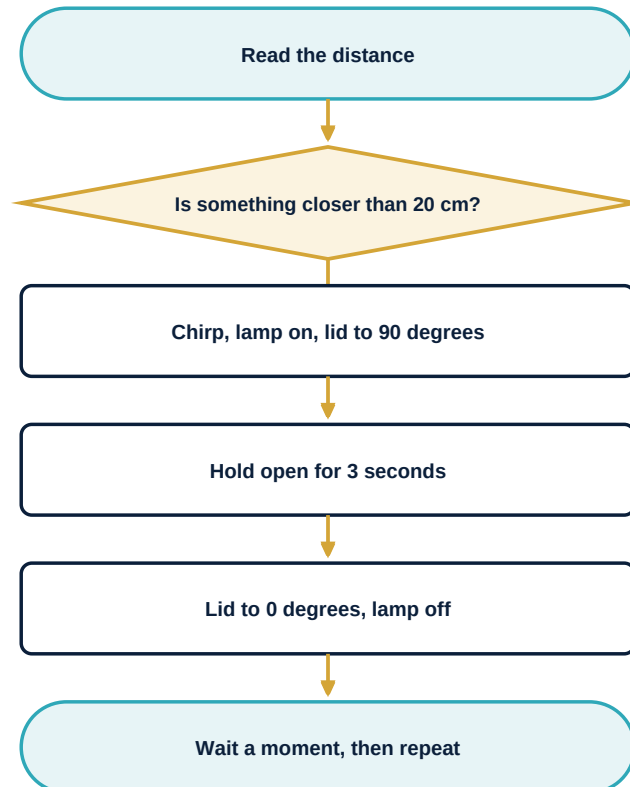


Figure 4.3 - The open-hold-close cycle, drawn before it is coded

SmartDustbin.ino

```

#include <Servo.h>

Servo lid;
int trig = 9, echo = 10, buzzer = 11, lamp = 12;
int openDistance = 20;      // centimetres - tune this yourself

void setup() {
  lid.attach(6);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(lamp, OUTPUT);
  lid.write(0);              // start closed
  Serial.begin(9600);
}

int readDistance() {
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
  long t = pulseIn(echo, HIGH);
  return t * 0.034 / 2;
}

void loop() {
  int d = readDistance();
  Serial.println(d);

  if (d > 0 && d < openDistance) {
    digitalWrite(lamp, HIGH);
    digitalWrite(buzzer, HIGH);
    delay(80);
    digitalWrite(buzzer, LOW);

    lid.write(90);           // open
    delay(3000);             // hold
    lid.write(0);            // close
    digitalWrite(lamp, LOW);
    delay(500);              // settle before looking again
  }

  delay(100);
}

```

Notice `d > 0` in the condition. When the sensor hears no echo at all, `pulseIn` eventually gives up and returns 0, which would otherwise look like *something is touching me* and fling the lid open at random. That one small test removes an entire class of mysterious faults.

SESSION 11

one class

False Triggers and How to Beat Them

Today you will

- Find out what makes the bin open when it should not.
- Test with real users, not just your own hand.
- Tune the threshold and the hold time.

Your bin works. Now find out how it **fails**, because a machine is only finished when you know its weaknesses. A trigger that happens when nobody meant it is called a **false trigger**, and every automatic machine in the world fights them.

What you see	Why it happens	What to do about it
Lid opens with nobody there	The sensor sees the wall, a chair or its own open lid	Aim it outwards and check nothing crosses its view
Lid opens twice for one hand	The hand is still there when the loop looks again	Increase the settle delay after closing
Lid opens very late	Threshold set too close to the bin	Raise openDistance towards 25 or 30
Lid trembles while open	The servo is straining against a heavy lid	Lighten the lid or move the horn closer to the hinge
Random opening at 0 cm	No echo returned and pulseIn gave up	Keep the d > 0 test in your condition

Now test with real users. Ask three classmates who have not seen your bin to throw a paper ball into it, and watch without saying anything. People approach at odd angles, from the side, or too fast. What you learn in those thirty seconds is worth more than an hour of adjusting numbers on your own.

Acceptance test	Result	Tick
Lid opens within 1 second of a hand at 20 cm	_____	■
Lid stays open about 3 seconds	_____	■
Lid closes fully every time, with no trembling	_____	■
Ten approaches in a row, no false trigger	_____	■
Left running for two minutes untouched: stays shut	_____	■

Challenge

Add a second HC-SR04 pointing down inside the bin to measure the fill level. When rubbish rises above a set line, flash the lamp to say the bin needs emptying. Real smart bins in airports do exactly this and send a message to the cleaning staff.

New word	What it means
Servo	A motor that turns to an exact angle and holds it.
Actuator	Any part that moves or acts: a motor, a buzzer, a lamp.
Horn	The plastic arm attached to a servo's shaft.
False trigger	The machine acting when nobody intended it to.